Opinion | Why China said ‘no’ to stripped-down Nvidia chips

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Jinghan (Michael) Zeng September 23, 2025

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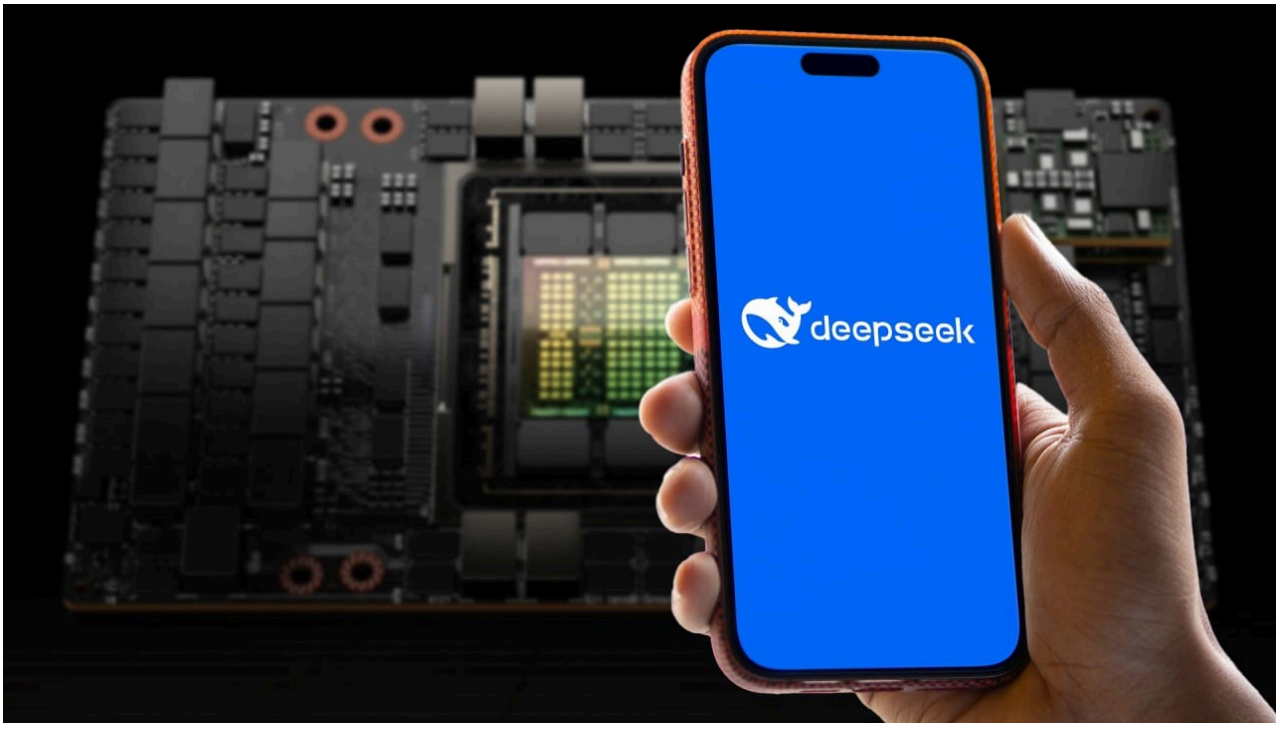
When US tech giant Nvidia unveiled its B40-series chips for China – stripped-down versions of its more advanced H20 line – many in Washington and Silicon Valley believed they had struck a clever compromise. The United States would retain its technological lead while allowing China to buy “safe” alternatives that preserved its dependence on American suppliers. Nvidia, in turn, could continue to profit from one of its largest markets.

The logic was simple: Beijing would accept something rather than nothing. Yet China has responded with a decisive “no”. Regulators have reportedly ordered leading technology companies to halt purchases of these chips. Beijing has also opened antitrust investigations into Nvidia and raised pointed questions about safety and security risks. What might look like a minor technical dispute is, in fact, a turning point in the politics of global technology.

For decades, Chinese technology development was built on global supply chains dominated by American firms. Reliance on US chips, software and designs was tolerated as the cost of participating in globalisation. After all, every country depends on others in some way. But Washington's decision to impose sweeping restrictions on high-end artificial intelligence (AI) chips shifted the calculus in Beijing.

By banning the most powerful versions while allowing watered-down ones like the B40, US regulators hoped to slow China's progress without forcing a full rupture. What they underestimated was how this policy reshaped debates inside China about technological dependence.
Once, many Chinese policymakers and entrepreneurs assumed that reliance on US hardware was inevitable – the unavoidable price of joining the global technology game. Building an entirely self-sufficient semiconductor industry seemed overwhelming. Experts routinely warned that catching up with America in high-end chips would take decades, perhaps generations.

But once Washington cut off access to the best technology, the conversation in Beijing shifted from whether self-reliance was desirable to whether it was unavoidable. Every ban, every tightened rule, every new licensing requirement strengthened the voices calling for indigenous innovation. What once looked like a distant dream suddenly became a national necessity.



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Does the arrival of China's low-cost DeepSeek mean the end of Nvidia's chip dominance?

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China's rejection of Nvidia's "castrated" chips is therefore both symbolic and strategic. Symbolically, it rejects the notion that Beijing should be satisfied with second-class products deliberately designed to limit its capabilities. To continue buying such chips would be seen as accepting the technological inferiority dictated by Washington.

Strategically, Chinese alternatives are catching up. Chinese media highlight that domestic chips are performing at least on a par with the downgraded Nvidia models. Whether that is fully accurate, the perception itself is powerful. If home-grown products are good enough, why remain dependent on foreign ones – especially those deliberately constrained to underperform?

Trust and security also matter. Allegations have surfaced in Chinese media about potential back doors or vulnerabilities in the cut-down Nvidia chips, including fears they could be remotely disabled. Nvidia denies this, but the broader point resonates: why should China build its future digital infrastructure on components designed abroad, especially by a country that openly treats it as a strategic rival?

Even if unfounded, the risk is real. Technological dependence in an age of great power politics is not merely an economic issue but a national security liability – a logic that also underpins Washington's decisions to block advanced AI chips to China, blacklist Huawei Technologies and pressure ByteDance over alleged security risks and potential back doors controlled by Beijing.

The implications are significant. For Nvidia, the immediate effect is lost sales. China is one of its largest markets, and even restricted products represented billions of dollars in revenue. For US firms more broadly, each rejected compromise deepens supply chain decoupling.

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China shows off latest AI innovations at international conference in Shanghai
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For China, the refusal is a boost to its domestic industry. Chipmakers that once feared they could never compete with Nvidia now find themselves with a protected market, regulatory support and political momentum. Far from crippling China's AI ambitions, US export controls have arguably catalysed them.

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China's AI ecosystem is already growing faster than many predicted. Domestic players like Huawei, Cambricon Technologies and Biren Technology are iterating rapidly. Producing chips at scale that can rival Nvidia's best remains an enormous challenge, especially amid shortages of critical equipment like extreme ultraviolet lithography machines. But the direction is unmistakable: what was once dependency is now defined as vulnerability, and overcoming it has become a political and strategic imperative.
For Washington, the lesson may be that trying to calibrate China's dependence through partial restrictions is a losing game. The very act of restriction drives the outcome it sought to avoid: China's rapid pursuit of self-reliance.

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How Hangzhou's 'Six Little Dragons' built a new Chinese tech hub

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The refusal to accept castrated chips is more than a procurement decision. It signals a fundamental shift in China's technological strategy. No longer content to participate in globalisation on terms set by others, Beijing is willing to bear the cost of building its own path.

That path may be bumpy and expensive in the short term. But in the long run, it could leave China with a technology base that is more resilient, more autonomous and embedded in a global network of Belt and Road Initiative partners that buy Chinese AI products.
Washington hoped to maintain China's dependence. Instead, it may have accelerated China's independence – and in the process, fuelled the rise of a parallel AI ecosystem that the world will now have to contend with.

Jinghan (Michael) Zeng

Jinghan (Michael) Zeng is a professor in the Department of Public and International Affairs at City University of Hong Kong. His research interests lie in politics and international relations, with a focus on China. His current work explores AI, the Belt and Road Initiative and Confucius Institutes. He is the author of "Memoirs of a Confucius Institute Director", "Artificial Intelligence with Chinese Characteristics: National Strategy, Security and Authoritarian Governance", among others.